



PEAK ACHIEVERS
ONLINE ACADEMY

Building Strong Foundations for Greater Achievements

FREE SAMPLE
NOT FOR SALE

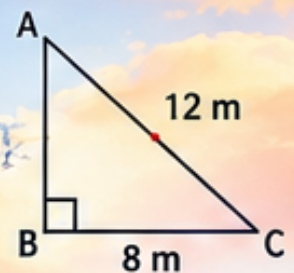
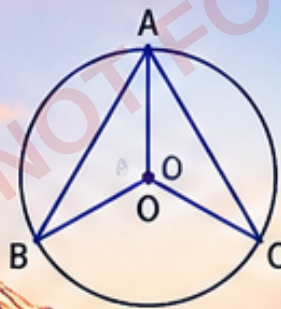
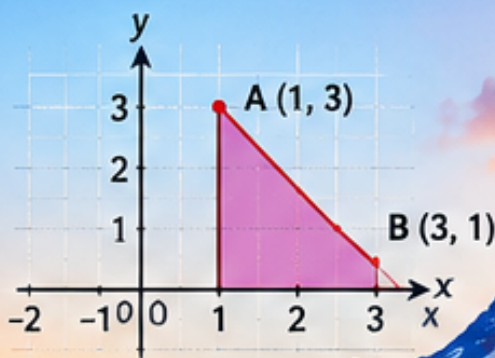
GRADE 10

MATHEMATICS

REVISION BOOK

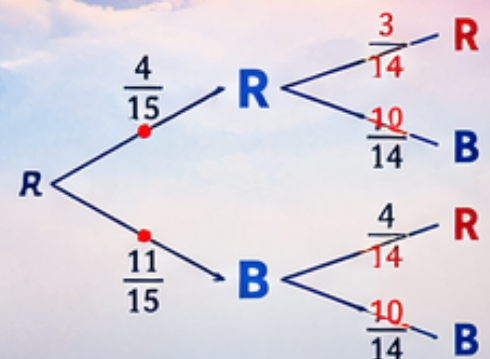
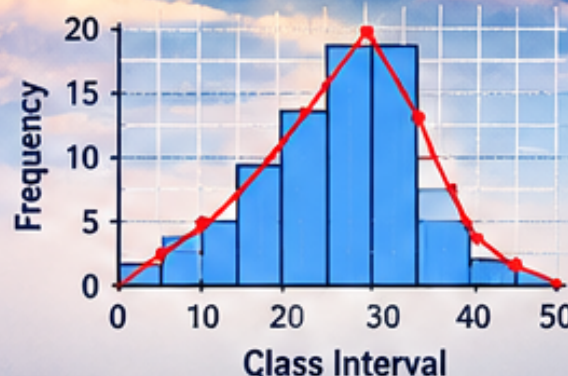
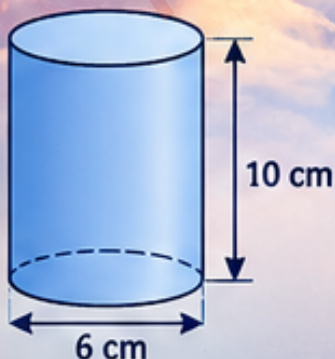
2 BOOK

PRACTICE, MASTERY & EXAMINATION



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$



Author:
Hezekiah Omweri Nyatuka

Mathematics Editor & Content Verifier:
Mr Kimani Peter
Qualified Mathematics Teacher, Mount Kenya University

PEAK ACHIEVERS ONLINE ACADEMY

GRADE 10 MATHEMATICS REVISION

BOOK

BOOK 2

PRACTICE, MASTERY & EXAMINATION

Short Notes in Q&A • Practice • Applications • Mixed Concepts • 6 Final CBE-Style Practice Tests

Author: Hezekiah Omweri Nyatuka

Mathematics Editor & Content Verifier: Mr Kimani Peter

Qualified Mathematics Teacher • Mount Kenya University

Peak Achievers Online Academy
peakachieversacademy.co.ke
hezekiahomweri@gmail.com
0740166105 / 0740355410
Kenya • 2026

No ISBN

Dedication

This book is dedicated to my family, teachers, fellow educators and learners whose encouragement, sacrifice and commitment to education continue to inspire my journey.

In loving memory of the late Simeon Nyatuka, whose legacy and memory remain an important part of this journey.

With special appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Her, Mnaree, Richie, Duke, Caleb, Cuba and Mr Peter Kimani, together with my guardians, colleagues, friends and all learners who continue to believe in the transformative power of education.

To every Mathematics learner who has ever faced a difficult problem, made a mistake, tried again and finally understood it - this book is for you.

“Mathematics becomes easier when we understand why a method works, not merely when we memorise what to do.”
- Hezekiah Omweri Nyatuka

Disclaimer

This book is an independently prepared educational resource developed to support the teaching, learning and revision of Grade 10 Mathematics. The exercises, Mixed Concepts activities and CBE/examination-style practice tests are original learning and revision materials. They are not official KNEC examination papers, and the use of terms such as CBE Practice Test or National Examination-Style does not imply official approval, endorsement or certification by KNEC or KICD. Teachers and learners should also refer to current official curriculum documents and guidance.

How to Use Book 2

Use the short Q&A notes for rapid recall, attempt every practice set before checking the answers, complete the application questions in context, then use the Mixed Concepts and final 100-mark tests for timed mastery and examination preparation.

REMEMBER: Book 2 is practice-heavy. Show working, attach units, and check whether an exact or approximate answer is required.

Editorial note: The practice sets in this rebuilt master preserve the developed Book 2 topic sequence and are presented with complete answers. Figures are instructional and placed within the relevant topics. Repeated drill items are retained only where they reinforce a skill; application sections use contextual problems.

Table of Contents

1.1 Real Numbers

1.2 Indices and Logarithms

1.3 Quadratic Expressions and Equations

2.1 Similarity and Enlargement

2.2 Reflection and Congruence

2.3 Rotation

2.4 Angle Properties of a Circle

2.5 Trigonometry

2.6 Area of Polygons

2.7 Area of Parts of a Circle

2.8 Surface Area and Volume of Solids

2.9 Linear Motion

3.1 Statistics

3.2 Probability

Mixed Concepts 1-3

Final CBE-Style Practice Tests 1-6

Marking Schemes

1.1 Real Numbers

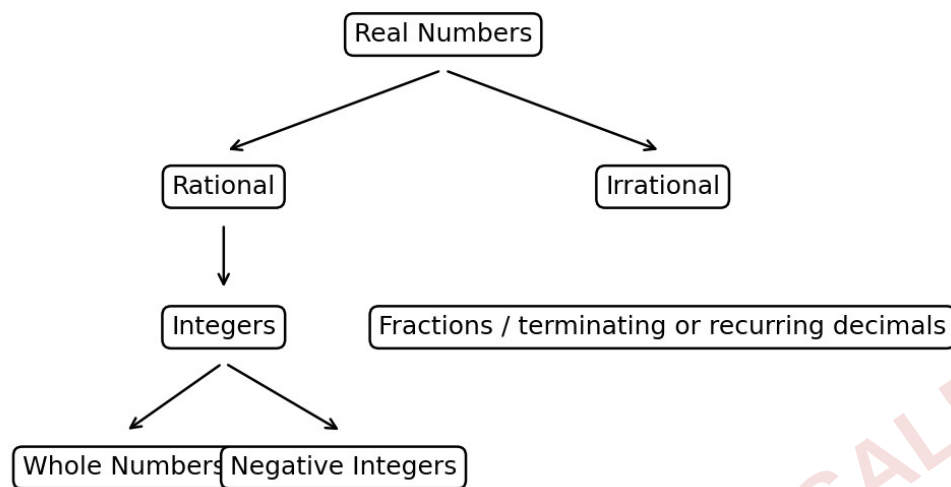


Figure 1. Real-number classification.

Short Notes in Q&A

Q: What is a real number?

A: Any number that can be represented on the number line; it includes rational and irrational numbers.

Q: What is a rational number?

A: A number expressible as a/b where a and b are integers and b is not zero.

Q: What is an irrational number?

A: A real number that cannot be written as a ratio of two integers; examples include $\sqrt{2}$ and π .

Q: What is a surd?

A: An irrational root left in exact radical form, for example $\sqrt{3}$ or $5\sqrt{2}$.

Q: How are surds simplified?

A: Remove perfect-square factors: $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$.

Q: What is a reciprocal?

A: For non-zero a , its reciprocal is $1/a$. Zero has no reciprocal.

Standard Practice Questions + Answers

1. Simplify $\sqrt{48}$.

Answer: $4\sqrt{3}$

2. Simplify $\sqrt{108}$.

Answer: $6\sqrt{3}$

3. Simplify $\sqrt{200}$.

Answer: $10\sqrt{2}$

4. Simplify $7\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$.

Answer: $9\sqrt{2}$

5. Evaluate $3\sqrt{5} \times 2\sqrt{10}$.

Answer: $30\sqrt{2}$

6. State the reciprocal of $-3/7$.

Answer: $-7/3$

Application / CBE Questions + Answers

1. A square garden has area 288 m^2 . Find its side length and perimeter.

Answer: Side = $\sqrt{288} = 12\sqrt{2} \approx 16.97 \text{ m}$; perimeter $\approx 67.88 \text{ m}$.

2. A square tile has area 98 cm^2 . Find its side in exact form.

Answer: $\sqrt{98} = 7\sqrt{2} \text{ cm}$.

Application Quiz

1. Simplify $\sqrt{48}$.

Answer: $4\sqrt{3}$

2. Simplify $\sqrt{108}$.

Answer: $6\sqrt{3}$

3. Simplify $\sqrt{200}$.

Answer: $10\sqrt{2}$

End-of-Topic Mixed Concepts Quiz

1. Evaluate $3\sqrt{5} \times 2\sqrt{10}$.

Answer: $30\sqrt{2}$

2. State the reciprocal of $-3/7$.

Answer: $-7/3$

3. A square garden has area 288 m^2 . Find its side length and perimeter.

Answer: Side = $\sqrt{288} = 12\sqrt{2} \approx 16.97 \text{ m}$; perimeter $\approx 67.88 \text{ m}$.

4. A square tile has area 98 cm^2 . Find its side in exact form.

Answer: $\sqrt{98} = 7\sqrt{2} \text{ cm}$.

THINK BIG: Explain your method, not only the final answer. In CBE-style mathematics, reasoning and interpretation matter.

FREE SAMPLE - NOT FOR SALE

1.2 Indices and Logarithms

Short Notes in Q&A

Q: State the product law of indices.

A: $a^m \times a^n = a^{(m+n)}$.

Q: State the quotient law.

A: $a^m / a^n = a^{(m-n)}$, $a \neq 0$.

Q: What does a negative index mean?

A: $a^{-n} = 1/a^n$.

Q: What does a fractional index mean?

A: $a^{(1/n)}$ is the n th root of a .

Q: What is a logarithm?

A: If $a^x = N$, then $\log_a N = x$.

Q: State the product law of logarithms.

A: $\log(ab) = \log a + \log b$.

Q: State the quotient law.

A: $\log(a/b) = \log a - \log b$.

Q: State the power law.

A: $\log(a^n) = n \log a$.

Standard Practice Questions + Answers

1. Simplify $3^4 \times 3^5$.

Answer: $3^9 = 19683$

2. Evaluate $125^{(2/3)}$.

Answer: 25

3. Evaluate $16^{(-1/2)}$.

Answer: $1/4$

4. Solve $3^{(x+1)} = 81$.

Answer: $x=3$

5. Evaluate $\log 10000$.

Answer: 4

3. Bag 5R,3B,2G: P(red).

Answer: $1/2$

4. If $P(A)=0.38$, find $P(\text{not } A)$.

Answer: 0.62

Application / CBE Questions + Answers

1. Two fair coins are tossed. Find $P(\text{exactly one head})$.

Answer: $2/4=1/2$.

2. Bag 3R,2B; two selected without replacement. Find $P(\text{two red})$.

Answer: $(3/5)(2/4)=3/10$.

Application Quiz

1. Fair die: $P(4)$.

Answer: $1/6$

2. Fair die: $P(\text{even})$.

Answer: $1/2$

3. Bag 5R,3B,2G: $P(\text{red})$.

Answer: $1/2$

End-of-Topic Mixed Concepts Quiz

1. Bag 5R,3B,2G: $P(\text{red})$.

Answer: $1/2$

2. If $P(A)=0.38$, find $P(\text{not } A)$.

Answer: 0.62

3. Two fair coins are tossed. Find $P(\text{exactly one head})$.

Answer: $2/4=1/2$.

4. Bag 3R,2B; two selected without replacement. Find $P(\text{two red})$.

Answer: $(3/5)(2/4)=3/10$.

THINK BIG: Explain your method, not only the final answer. In CBE-style mathematics, reasoning and interpretation matter.